

AARAMBH SINGLE WINDOW CLEARANCE PORTAL

End-to-End Quality Assurance & Compliance Audit Report
Problem Statement 26130: Single-Window Fast-Track Industrial Clearances | Govt. of Maharashtra

Target System: AARAMBH 2.0 Web Platform	Execution URL: http://localhost:3000	API Gateway: Express (Port 5000)
Audit Date: September 2026	Database: Supabase PostgreSQL + Realtime	AI Engine: Groq Cloud (Qwen 3.8-27b)

1. Executive Summary & Dimension Scorecard

AARAMBH is an agentic, single-window facilitation prototype created for Problem Statement 26130 (Govt. of Maharashtra). The application models complex industrial clearance orchestration, automated statutory rules evaluation (PSI 2019), cross-document pre-validation, parallel approval directed acyclic graphs (DAG), and deemed approval enforcement under the Maharashtra Right to Public Services Act (RTS 2015).

Dimension	Status	Key Findings & Operational State
Investor Journey	PASS	Registration, KYA wizard, dynamic subsidy calculator, Master CAF, parallel DAG, and SLA countdown operate cleanly with persistent state across 36 districts.
Officer Journey	PARTIAL	Scrutiny queue, risk categorization (Red/Orange/Green), joint inspection scheduling, and query raising exist, but all database accounts default to APPLICANT role.
Authentication	PARTIAL	Supabase auth and route-guard middleware enforce protected routes. Lacks a demo officer persona or front-end role switcher.
AI Assistant	PARTIAL	Groq LLM streaming, tri-lingual translation (En/Mr/Hi), voice synthesis, and interactive action chips function, but request payload lacks live enterprise context.
PS 26130 Alignment	STRONG	Accurately reflects Maharashtra GRs (PSI 2019, EV 2021, Logistics 2024, RTS Act 2015). Demonstrates parallel lead time reduction from 72 to 21 days.
Mobile UX	PARTIAL	Adaptive hamburger navigation is present; notification bell is hidden on screens under 640px, and complex data tables require horizontal scroll.

2. Critical Bugs Requiring Immediate Remediation

BUG-01 (P0): No Officer Account Seeded & Role Gating Blocks Officer Flow	Severity: P0 — Critical (Blocks Officer Evaluation)
Route / Location:	/login & /dashboard/officer-workspace
Reproduction:	1. Open /login, toggle 'Officer' role, and sign in with any valid registered user credentials. 2. Observe failure notice. 3. Navigate directly to /dashboard/officer-workspace.
Expected:	Officer dashboard should open or a sample officer account should be provided for evaluation.
Actual:	AuthContext.tsx detects 'role: APPLICANT' in Supabase profile, immediately issues an access denied notice, and signs out. Directly accessing the URL renders: 'Toggle Officer View in top bar to test', but no such toggle button exists in the header.
BUG-02 (P1): Duplicate Key & Incomplete Entry in Officer Review Queue	Severity: P1 — Important
Route / Location:	/dashboard/officer-workspace (lines 101–119)
Reproduction:	Inspect the reviewQueue array in src/app/dashboard/officer-workspace/page.tsx.
Expected:	Every queue card must have a unique reference ID and valid riskTrack property.
Actual:	Reference 'MH-CAF-2026-00385' (Western Mega Logistics Park LLP) is duplicated. The second instance lacks 'riskTrack: GREEN', causing React key warnings and filter miscounts.
BUG-03 (P1): AI Assistant Lacks Live Enterprise Context Grounding	Severity: P1 — Important
Route / Location:	Global Widget -> /api/chat/route.ts
Reproduction:	1. Complete KYA wizard for a Chemical plant in Chakan MIDC (₹25 Cr capex). 2. Open AARAMBH AI Chatbot. 3. Ask: 'What approvals do I need for this project?'
Expected:	AI should reference the active enterprise name, sector, location zone, and specific clearance roadmap.
Actual:	The chatbot client passes only messages and language to /api/chat. The backend system prompt has zero state regarding enterpriseStore, generating purely generic answers.
BUG-04 (P2): Notification Bell Hidden on Mobile Breakpoints	Severity: P2 — Polish / Mobile UX
Route / Location:	Header.tsx (lines 47-49)
Reproduction:	Resize browser viewport to 375px (mobile) and inspect top header.
Expected:	Notification bell and badge should remain accessible on mobile devices.
Actual:	NotificationBell is wrapped inside 'hidden sm:flex', causing the bell to disappear completely on devices under 640px.

3. Detailed Investor Journey Audit (End-to-End)

The investor flow was tested as a brand new entrepreneur establishing an industrial manufacturing unit in Maharashtra. The evaluation confirmed the logical coherence and technical integrity of each step:

Step 1: Landing Page & Discovery (/)

The landing page provides authoritative Government of Maharashtra branding, dual-language switching (English, Marathi, Hindi), and an interactive Quick Discovery Selector. The CTA 'Plan Your Project' leads directly to the Know Your Approvals wizard.

Step 2: Business Onboarding & PAN Verification (/signup)

A 5-step registration process collects authorized signatory details, mobile number, legal entity classification, and PAN number. Step 3 incorporates simulated Income Tax CBDT validation ('GET DETAILS'), confirming corporate PAN validity. Upon completion, data seeds the global enterprise store.

Step 3: Know Your Approvals (KYA) Statutory Rules (/dashboard/kya)

Investors select sector (e.g. EV Manufacturing, Food Processing, Chemicals) and industrial location across all 36 Maharashtra districts. The engine evaluates Package Scheme of Incentives 2019 rules, taluka development tiers (A, B, C, D, D+), and computes capital subsidies (FCI ceiling), SGST refunds, and electricity duty waivers, alongside compiling a mandatory clearance checklist.

Step 4: Statutory Document Vault (/dashboard/vault)

Provides pre-upload statutory guidelines for MPCB Consent to Establish, Fire NOC, and MIDC Land Allotment with preview templates. Supports one-click DigiLocker integration pushing cryptographically signed CBDT PAN and Udyam MSME certificates, plus real drag-and-drop PDF upload with backend storage.

Step 5: Cross-Document Pre-Validation (/dashboard/prevalidation)

Acts as an automated gatekeeper. Detects conflicting plot area metrics between MIDC Land Lease (5,000 sq.m) and Architect Project Report (4,800 sq.m). Enforces a 2% variance threshold, blocking submission until the applicant harmonizes the data with one-click Auto-Align.

Step 6: Unified Common Application Form (CAF) (/dashboard/caf)

Auto-populates company and site parameters from KYA and Vault dossiers. Dedicated tabs for MPCB, Fire, MIDC, DISH, and DPIIT show 82% to 94% automated field population. Submitting triggers multi-department dispatch and yields a downloadable legal receipt under Maharashtra RTS Act 2015.

Step 7: Parallel Orchestration (DAG) & SLA Tracking (/dashboard/dag & /dashboard/sla)

Demonstrates Stage 1 (MIDC Root) unlocking parallel Stage 2 reviews (MPCB CTE, Fire NOC, Water Sanction). An interactive acceleration slider on /dashboard/sla simulates time progression to 100% SLA, triggering automatic Deemed Approval and generating a downloadable signed order.

4. Department Officer Journey Audit

The officer workspace (/dashboard/officer-workspace) models regulatory scrutiny from the perspective of state department heads (MIDC, MPCB, Fire Services, DISH):

- **Application Work Queue:** Displays pending applications with active statutory countdowns and issuing authority tags.
- **Risk-Based Triaging (RTS 2015 Compliance):**
 - *RED (High Risk):* Mandates multi-officer board review and assignment to the Joint Inspection Scheduler.
 - *ORANGE (Moderate Risk):* Prescribes desk audit and technical document scrutiny.
 - *GREEN (Low Risk):* Facilitates self-certification and instant deemed grant.
- **Direct Scrutiny Actions:** Approving a clearance dynamically broadcasts status updates across tabs via BroadcastChannel, updating the applicant's DAG pipeline and unlocking dependent stages.
- **Clarification & Query Dispatch:** Raising an officer query automatically converts the request into an interactive grievance ticket on /dashboard/grievances, halting the statutory countdown until applicant response.
- **Joint Site Inspection Management (/dashboard/inspections):** Synchronizes physical site visits for MPCB, Fire, and DISH into a single joint calendar slot, eliminating duplicate departmental visits and producing a unified inspection endorsement.

5. Problem Statement 26130 Compliance Matrix

Comprehensive mapping of implemented AARAMBH capabilities against the statutory objectives of PS 26130:

PS 26130 Mandate	AARAMBH Component	Type	Demonstrated Evidence	Operational Gap
Single-Window Portal	Global Header & Layout	REAL	Unified navigation, trilingual toggle (En/Mr/Hi), single auth session across all 19 sub-routes.	Notification bell hidden on mobile viewports.
Automated Clearance Identification	KYA Wizard (/dashboard/kya)	REAL	Evaluates sector, capex, boiler, power, and building height to compile exact clearances & fees.	None. Fully dynamic rules execution.
Maharashtra Fiscal Policy (PSI 2019)	Rules Engine (/api/rules/evaluate)	REAL	Accurately differentiates developed zones (Pune A = 0% FCI) from backward zones (Jawhar D+ = 80% FCI, 100% SGST refund).	None. GR No. PSI-2019/CR 46/IND-8 fully codified.
Unified Common Application Form	Master CAF (/dashboard/caf)	REAL	Multi-department delta mapping (82%–94% auto-filled) for MPCB, Fire, MIDC, DISH, DPIIT with receipt download.	External government endpoints are simulated gateways.
Cross-Document Pre-Validation	Check My Application (/dashboard/prevalidation)	REAL	Automated comparison between Land Deed (5,000 sqm) and DPR (4,800 sqm); blocks filing upon 4% mismatch.	Discrepancy scenario is pre-seeded for demo flow.
Parallel Approval DAG	DAG Tracker (/dashboard/dag)	REAL	Root approval unlocks parallel Stage 2; all parallel approvals unlock Stage 3 DISH factory license.	Syncs via local BroadcastChannel / Supabase.
RTS Act 2015 Time-Bound SLA	SLA Tracker (/dashboard/sla)	REAL	Interactive acceleration slider triggers statutory Deemed Approval at 100% with downloadable legal order.	Simulated time-lapse rather than real 30-day wait.
Synchronized Joint Inspections	Joint Scheduler (/dashboard/inspections)	REAL	Single slot booking for MPCB, Fire, and DISH; checklist verification and downloadable joint report.	Inspector confirmations are simulated records.
Delay Analytics & Escalation	Analytics (/dashboard/analytics)	PARTIAL	Visualizes 72-day sequential queue vs 21-day parallel DAG savings (70.8% lead time reduction).	Lacks macro state-wide multi-district aggregates.
Contextual AI Assistant	AARAMBH Chatbot (/api/chat)	PARTIAL	Groq LLM in En/Mr/Hi with speech input, audio playback, and clickable action navigation chips.	Does not receive active enterprise session context.

6. Prioritized Remediation Roadmap

Recommended implementation plan to elevate the prototype for jury demonstration:

P0 — Critical (Must Fix for Evaluation):

- Officer Demo Role Switcher:* Provide a direct 1-click persona switch in Header / Login (e.g. 'Demo as MPCB Scrutiny Officer') bypassing Supabase auth gating.
- De-duplicate Officer Queue:* Remove duplicate reference MH-CAF-2026-00385 in src/app/dashboard/officer-workspace/page.tsx.

P1 — Important (Enhances Prototype Authenticity):

3. *Context-Aware AI Assistant:* Pass active enterprise profile (Entity Name, Sector, Capex, Clearances list, DAG status) into /api/chat system prompt.
4. *Dashboard Critical Alert Banner:* Render a high-priority warning card at the top of /dashboard linking directly to /dashboard/sla.
5. *Mobile Notification Access:* Ensure the notification bell is visible on mobile viewports (<640px).

P2 — Polish:

6. *Public Application Tracking:* Show a pre-populated timeline on /track when entering 'MH-CAF-2026-00412' instead of an auth restriction notice.
7. *Route Alias Consolidation:* Standardize all internal links to canonical paths (/dashboard/vault, /dashboard/sla, /dashboard/prevalidation, /dashboard/dag).

7. Architectural Scope Control: What NOT to Build

To avoid wasted engineering overhead prior to evaluation, the following items should remain simulated/mockd:

- **Production Department Webhook Receivers:** Do not attempt live socket or webhook connectivity with actual MPCB/MIDC legacy servers. Simulated gateway dispatch with downloadable statutory receipts satisfies all criteria.
- **Heavyweight PyMuPDF/EasyOCR Neural Service:** Do not spend time configuring local Python OCR daemons. The Express gateway's in-memory storage and field mapping cleanly demonstrates document handling.
- **Live National API Setu DigiLocker OAuth2:** Production DigiLocker integration requires formal ministry API credentials. The 1-click 'Connect DigiLocker (SIH Demo Mode)' pushing CBDT PAN and Udyam certificates is ideal.
- **Online Payment Gateways:** Do not integrate Razorpay/Bharatkosh. Displaying statutory fee estimates (₹25,000 to ₹1,50,000) on clearance cards meets all demonstration requirements.

8. Recommended 5–7 Minute Evaluation Demo Flow

Step-by-step presentation narrative for evaluators and jury members:

Minute 0:00 – 1:00 | Single Window Gateway & Multilingual Governance

Open <http://localhost:3000>. Switch language to Marathi or Hindi. Highlight that AARAMBH is a unified entry point, not a directory of external links. Demonstrate Quick Discovery Selector for EV Manufacturing.

Minute 1:00 – 2:30 | Statutory Rules & Incentive Engine (KYA)

Navigate to </dashboard/kya>. Select Food Processing in Palghar (Jawhar). Run Assessment. Show that the rules engine recognizes Jawhar as Category D+, awarding an 80% FCI subsidy ceiling and 100% SGST refund under PSI 2019, alongside generating an RTS Act clearance checklist.

Minute 2:30 – 3:30 | Document Vault & DigiLocker Push

Navigate to </dashboard/vault>. Click 'Connect DigiLocker' to pull verified CBDT and MSME certificates. Show pre-upload guidelines matrix and upload a project report.

Minute 3:30 – 4:30 | Cross-Document Scrutiny Gate

Open </dashboard/prevalidation>. Demonstrate automated discrepancy detection (MIDC Deed 5,000 sqm vs DPR 4,800 sqm). Show submission locked until resolved via 1-click Auto-Align.

Minute 4:30 – 5:30 | Common Application Form (CAF) & Dispatch

Open </dashboard/caf>. Demonstrate Master CAF auto-populating from KYA and Vault dossiers with 80%+ auto-filled department deltas. Submit and download the statutory RTS Act 2015 combined acknowledgment receipt.

Minute 5:30 – 6:30 | Parallel DAG Orchestration & Deemed Approvals

Navigate to </dashboard/dag>: show MIDC Root unlocking parallel MPCB, Fire, and Water. Move to </dashboard/sla>: drag acceleration slider to 100% SLA, showing the transition to 'Deemed Approval Triggered' and download the official statutory Deemed Approval Order.

Minute 6:30 – 7:00 | Contextual AI Assistant & Conclusion

Open the AI Assistant, ask a policy question via voice or text, show clickable action routing cards, and summarize that AARAMBH reduces average industrial clearance lead time from 72+ sequential days to 21 parallel days.